

Supernetting

Task 1

Supernet the subnets
Find the new subnet mask

All of the below are networks

Which can be supernetted together? (there cannot be any gaps)

What would the new subnet mask be?

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

192.168.0.0

192.168.0.4

192.168.0.8

192.168.0.12

All of the below are networks

Which can be supernetted together? (there cannot be any gaps)

What would the new subnet mask be?

start

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>		<u>2</u>	<u>1</u>
7	6	5	4	3	2		1	0

They're going up by 4s

Group by 4

192.168.0.0

192.168.0.4

192.168.0.8

192.168.0.12

192.168.0.0 /28 255.255.255.240

Final
Result!

Task 2

Supernet the subnets
Find the new subnet mask

All of the below are networks with the same subnet mask
Which can be supernetted together? (there cannot be any gaps)
What would the new subnet mask be?

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

192.168.0.16
192.168.0.20
192.168.0.24
192.168.0.28
129.168.0.36

All of the below are networks with the same subnet mask
 Which can be supernetted together? (there cannot be any gaps)
 What would the new subnet mask be?

super				start			
<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

They're going up by 4s

Group by 4

192.168.0.16	}	192.168.0.16 /28 255.255.255.240
192.168.0.20		
192.168.0.24		
192.168.0.28		
192.168.0.36	←	Remaining 192.168.0.36 /252

Final
Result!

Task 3

Supernet the subnets
Find the new subnet mask

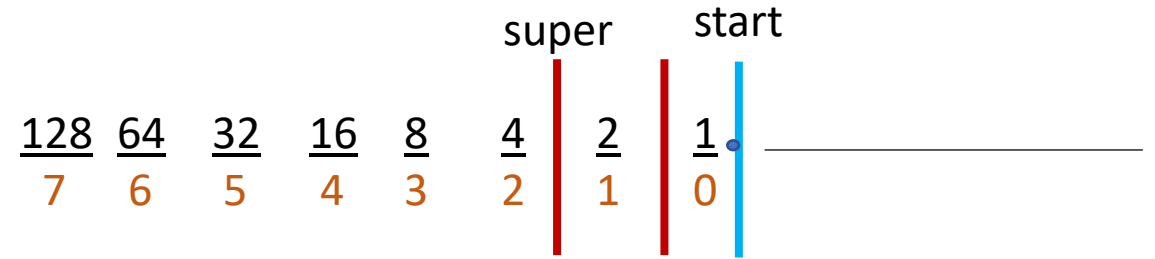
All of the below are networks with the same subnet mask
Which can be supernetted together? (there cannot be any gaps)
What would the new subnet mask be?

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

192.168.1.0
192.168.2.0
192.168.3.0
192.168.4.0
192.168.5.0
192.168.6.0
192.168.7.0
192.168.8.0

All of the below are networks with the same subnet mask
 Which can be supernetted together? (there cannot be any gaps)
 What would the new subnet mask be?

- They're going up by 1s
- Starting subnet mask is /24
- There are 8 of them
- Starting network must be an even number
- Deltas could be 2 or 4



192.168.1.0	_____	192.168.1.0 /24	255.255.255.0
192.168.2.0	}	192.168.2.0 /23	_____
192.168.3.0			
192.168.4.0	}	192.168.4.0 /23	_____
192.168.5.0			
192.168.6.0	}	192.168.6.0 /23	_____
192.168.7.0			
192.168.8.0	_____	192.168.8.0 /24	255.255.255.0

**Final
Result!**

Task 4

Supernet the subnets
Find the new subnet mask

All of the below are networks – their subnet masks are variable

Which can be supernetted together? (there cannot be any gaps)

What would the new subnet mask be?

100.100.100.0

100.100.100.8

100.100.100.16

100.100.100.24

100.100.100.32

100.100.100.40

100.100.100.48

100.100.100.56

100.100.100.64

100.100.100.68

100.100.100.72

100.100.100.76

100.100.100.88

100.100.100.92

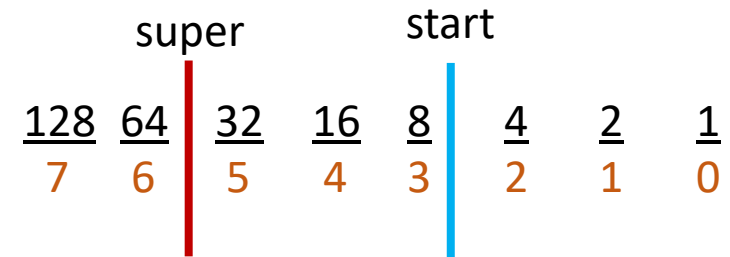
<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

All of the below are networks – **their subnet masks are variable**
Which can be supernetted together? (there cannot be any gaps)
What would the new subnet mask be?

100.100.100.0
100.100.100.8
100.100.100.16
100.100.100.24
100.100.100.32
100.100.100.40
100.100.100.48
100.100.100.56
100.100.100.64
100.100.100.68
100.100.100.72
100.100.100.76
100.100.100.88
100.100.100.92

- The first 8 are going up by 8s
- Starting delta 8
- Can group the first 8 together
- Supernet delta is 64 (8 x 8)
- Supernet IP range is .0 - .63

100.100.100.0 /26 255.255.255.192



All of the below are networks – **their subnet masks are variable**
 Which can be supernetted together? (there cannot be any gaps)
 What would the new subnet mask be?

100.100.100.0
 100.100.100.8
 100.100.100.16
 100.100.100.24
 100.100.100.32
 100.100.100.40
 100.100.100.48
 100.100.100.56
 100.100.100.64
 100.100.100.68
 100.100.100.72
 100.100.100.76
 100.100.100.88
 100.100.100.92

100.100.100.0 /26 255.255.255.192

- These are going up by 4s
- Starting delta 4
- Can group 4 together
- Supernet delta is 16 (4 x 4)
- Supernet IP range is .64 – .79
- 100.100.100.64 /28 255.255.255.240

super			start		
<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>
7	6	5	4	3	2

All of the below are networks – **their subnet masks are variable**
Which can be supernetted together? (there cannot be any gaps)
What would the new subnet mask be?

100.100.100.0
100.100.100.8
100.100.100.16
100.100.100.24
100.100.100.32
100.100.100.40
100.100.100.48
100.100.100.56

100.100.100.0 /26 255.255.255.192

100.100.100.64
100.100.100.68
100.100.100.72
100.100.100.76

100.100.100.64 /28 255.255.255.240

100.100.100.88
100.100.100.92

There are not enough remaining subnets to group together
With a delta of 4, the subnets go up by 4s
You would need 4 subnets to make a group

start

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

All of the below are networks – **their subnet masks are variable**
Which can be supernetted together? (there cannot be any gaps)
What would the new subnet mask be?

100.100.100.0	{	100.100.100.0 /26 255.255.255.192
100.100.100.8		
100.100.100.16		
100.100.100.24		
100.100.100.32		
100.100.100.40		
100.100.100.48	{	100.100.100.64 /28 255.255.255.240
100.100.100.56		
100.100.100.64		
100.100.100.68		
100.100.100.72	{	100.100.100.88 /30 255.255.255.252
100.100.100.76		
100.100.100.88	—	100.100.100.92 /30 255.255.255.252
100.100.100.92	—	

**Final
Result!**

